ORACLE SUPPORT SERVICES SUPPLEMENTAL GUIDE

The materials presented in this supplemental guide provide specific information for OUM practitioners to provide the guidance needed to effectively manage and support the lifecycle of Oracle environments during an implementation and after go-live. The Oracle Support Services Supplemental Guide contains the following components:

Oracle Support Services Lifecycle Management Strategy Overview

This component provides an overview of Oracle Support Services lifecycle management strategy along with an overview of the Information Technology Infrastructure Library (ITIL) Service Lifecycle upon which it is based. ITIL® is a Registered Trade Mark of the Office of Government Commerce in the United Kingdom and other countries.

Oracle Support Services Lifecycle Management Methodology Mapping

This component provides a mapping between the OUM and ITIL lifecycle management methodologies. The methodology mapping should be used to gain an understanding of the relationship between OUM and ITIL, as well as how to leverage the value of the ITIL best practices to achieve excellence in the lifecycle management of any Oracle investment.

Oracle Support Services Supplemental Task Guidelines for Lifecycle Management of the My Oracle Support Services Portal

This component should be used in conjunction with the standard OUM task guidelines to supplement baseline guidance for affected tasks when planning and implementing the processes, policies and procedures used for lifecycle management of the My Oracle Support Services portal.

Oracle Support Services Supplemental Task Guidelines for IT Change Management

This component should be used in conjunction with the standard OUM task guidelines to supplement baseline guidance for affected tasks when planning and implementing the processes, policies and procedures used for IT Change Management in Oracle environments.

Oracle Support Services Supplemental Task Guidelines for Incident Management

This component should be used in conjunction with the standard OUM task guidelines to supplement baseline guidance for affected tasks when planning and

implementing the processes, policies and procedures used for product support of your deployed applications. It focuses on Incident Management goals, processes and integration with your other IT service operations.

PDF View

You may also prefer to access this information through the **Oracle Support Services Supplemental Guide (PDF View)** located in the Supplemental Guides section of the Supplemental Guidance page.

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PURPOSE

The purpose of this document is to provide specific information for OUM practitioners regarding how to prepare customers to adopt and adapt established best practice guidance for effectively managing and supporting the lifecycle of Oracle environments during an implementation and after go-live.

This supplemental guide has been expanded for OUM 5.6 to include IT change management best practices in an Oracle Environment, as well as best practices contributing to the lifecycle management of the My Oracle Support Services portal.

My Oracle Support is an integral part of Oracle Customer Services' ongoing commitment to customer and partner success. The My Oracle Support application platform uses personalized, proactive, and collaborative capabilities to manage the lifecycle of support, including access to knowledge base articles, proactive and preventive services, and automated support tools.

Change management is a key process for keeping systems healthy. This enhanced version of the supplemental guide provides Oracle best practices related to people, process and technology in order to facilitate efficient and effective management of IT changes.

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AUDIENCE

This document is intended for OUM practitioners who will be preparing customers to adopt and adapt established best practice guidance for effectively managing and supporting the lifecycle of Oracle environments during an implementation and after go-live.

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HOW TO USE THIS DOCUMENT

This document contains supplemental guidance for planning and executing Oracle Support Services and should be used in conjunction with and not instead of the OUM base method guidance.

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LIMITATIONS TO THIS DOCUMENT

This document does not describe OUM in detail – the reader must review the documentation and training materials referenced below to gain a basic understanding of each.

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RELATED DOCUMENTATION AND TRAINING

The following related documentation and training provide additional detailed guidance related to OUM implementations and are prerequisites to this document:

- OUM Method Pack – Desktop Installable or Hosted version
- OUM Training and Accreditation Program

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ORACLE SUPPORT SERVICES OVERVIEW

INTRODUCTION

Successful management of an Oracle software and system investment from inception through retirement requires a lifecycle management strategy. The lifecycle management of Oracle software requires individuals that understand the technology and have the necessary skills to successfully manage the system throughout the lifecycle. Such knowledge is acquired through both training and experience.

An effective IT organization aligns itself with the business that it supports by delivering an optimal level of service at a justifiable cost. Achieving this level of best practice requires a framework on which the IT organization can build. Standards and methodologies provide repeatable and predictable means of delivering quality. Oracle-specific best practices, combined with industry standard frameworks, augment Oracle technology to deliver consistency and quality across the lifecycle. The Information Technology Infrastructure Library (ITIL) is an example of such an industry standard framework of best practices and is a starting point for defining and optimizing IT processes emphasizing service quality and continuous improvement. ITIL® is a Registered Trade Mark of the Office of Government Commerce in the United Kingdom and other countries.

ITIL was conceived in the late 1980s when the British Central Computer and Telecommunication Agency (CCTA), now called the Office of Government Commerce (OGC), made the determination that there should be a better way for Information Technology to function. The CCTA commissioned a study group to develop a new approach to managing Information Technology. From this group came Version 1 of ITIL, which was known as the Government Information Technology Infrastructure Management, or GITIM. Version 1 of ITIL was significantly different from the present-day version 3. Some of this difference is due to the gradual maturing of ITIL, while some is due to changes in the Information Technology industry.

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ITIL V3 SERVICE LIFECYCLE

The ITIL lifecycle consists of 5 phases:

- Service Strategy
- Service Design
- Service Transition
- Service Operations
- Continual Service Improvement

Service Strategy provides the framework for the tactical and strategic activities.

Service Design provides a framework for specifying requirements from the perspective of the customer defining how the requirements are fulfilled.

Service Transition manages the transition of new or changed services and IT management processes to production.

Service Operations is concerned with the day-to-day operations of services and IT management processes in support of the business.

Continual Service Improvement underpins all lifecycle phases and interacts with each phase in an ongoing manner asking, “What can we do better?”

The strength of this lifecycle approach is the continual feedback loop that runs through each phase. Feedback ensures that communication between each phase is constant, which provides opportunities for improvement. Continual evaluation and measurement of delivered services helps to validate the value provided, and ensures that the needs of the business are being addressed. Service Design, Service Transition and Service Operations implement the Service Strategy and are progressive phases of the lifecycle that represent change and transformation. Continual Service Improvement represents learning and improvement. Continual Service Improvement helps place and prioritize improvement programs and projects based on strategic objectives. Once again, ITIL ® is a Registered Trade Mark of the Office of Government Commerce in the United Kingdom and other countries. To learn more about ITIL, please visit the OGC Official Best Practice Site.

Oracle Customer Services has developed a complete portfolio of lifecycle services leveraging the ITIL framework. Please see the Oracle Customer Services Catalog for more information.

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REFERENCES

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ORACLE SUPPORT SERVICES LIFECYCLE MANAGEMENT METHODOLOGY MAPPING

INTRODUCTION

OUM incorporates and extends a number of industry standards including the Unified Process (UP), the Project Management Institute's Project Management Body of Knowledge (PMI PMBOK), the Unified Modeling Language (UML) and Business Process Modeling Notation (BPMN). With the introduction of the Support Services Supplemental Guide, OUM now incorporates best practices pertaining to service support as embodied in the Information Technology Infrastructure Library (ITIL).

OUM and ITIL leverage the value of best practice frameworks to achieve excellence in the lifecycle management of any Oracle investment. The ITIL framework is business driven and addresses the way an IT organization operates defining what an IT organization should do. OUM is a project management framework defining how to successfully implement Oracle solutions. ITIL answers the question "Are we doing the right things?" OUM answers the question "Are we doing things the right way?"

The focus of ITIL V3 is creating, delivering and managing the lifecycle of IT services and requires project and program planning for success. OUM includes three focus areas – Manage, Envision, and Implement. OUM's Manage focus area provides a framework in which all types of projects can be planned, estimated, controlled, and completed in a consistent manner. OUM's Envision focus area deals with development and maintenance of enterprise level IT strategy, architecture, and governance. Envision also assists in the transition from enterprise-level planning and strategy activities to the identification and initiation of specific projects. The Implement focus area provides a framework for developing and implementing Oracle solutions. Within ITIL, IT service delivery can be considered a project. Service as a project can be demonstrated as follows:

- A project has a beginning and an end. A service passes through phases during its lifecycle beginning with the requirements received and ending with delivery to the customer.
- The deliverable of a project is unique and is satisfied when the IT service is delivered to a specific customer in response to specific needs.
- The project execution is iterative and continually elaborating in response to changes which are satisfied through the ITIL continual service improvement and service transition processes.

The ITIL Service Portfolio contains the status of all services that are currently offered, have been offered in the past, and those that are 'pipe dreams,' 'nice to have' or ideas for the future. The Portfolio is comprised of three sections:

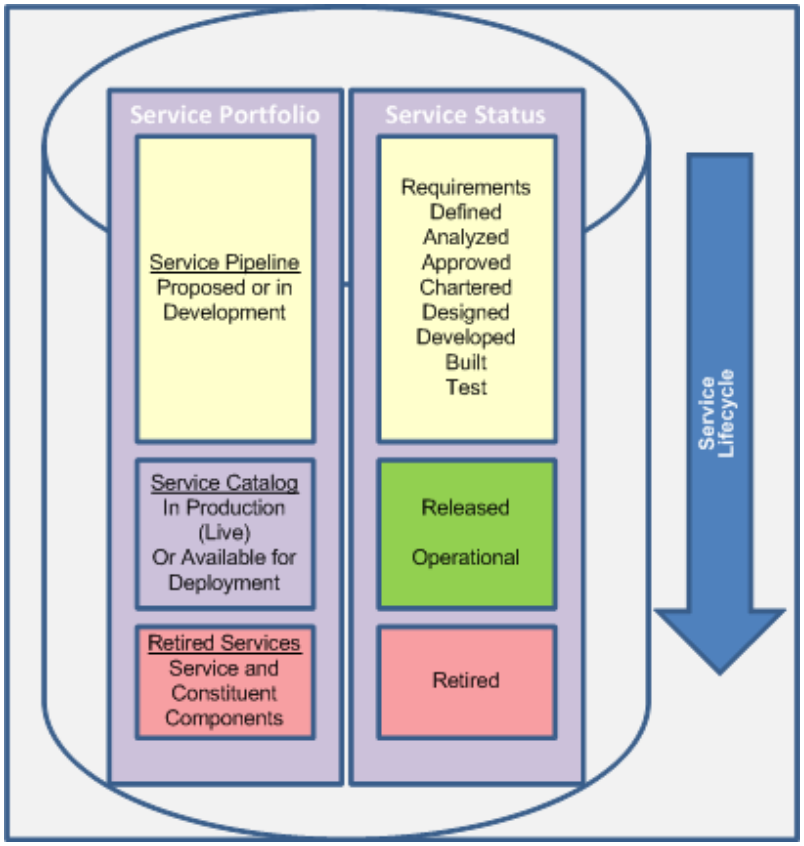
- Service Pipeline
- Service Catalog
- Retired Services

As a service progresses through the Service Lifecycle it is allocated a relevant status. The statuses a service may be assigned include:

- **Requirements** - set of outline requirements received from the business / IT for a new or changed service
- **Defined** - set of requirements for a new service are being assessed, defined and documented and the Service Level Requirements (SLRs) being produced
- **Analyzed** - set of requirements for a new service are being analyzed / prioritized
- **Approved** - set of requirements for a new service being finalized / authorized
- **Chartered** - new service requirements are being communicated, resources and budget allocated
- **Designed** - the new service and its constituent components are being designed and procured as required
- **Developed** - the service and its constituent components are being developed or harvested as required
- **Built** - the service and its constituent components are being built
- **Test** - the service and its constituent components are being tested
- **Released** - the service and its constituent components are being released
- **Operational** - the service and its constituent components are operational within the live environment
- **Retired** - the service and its constituent components are being retired

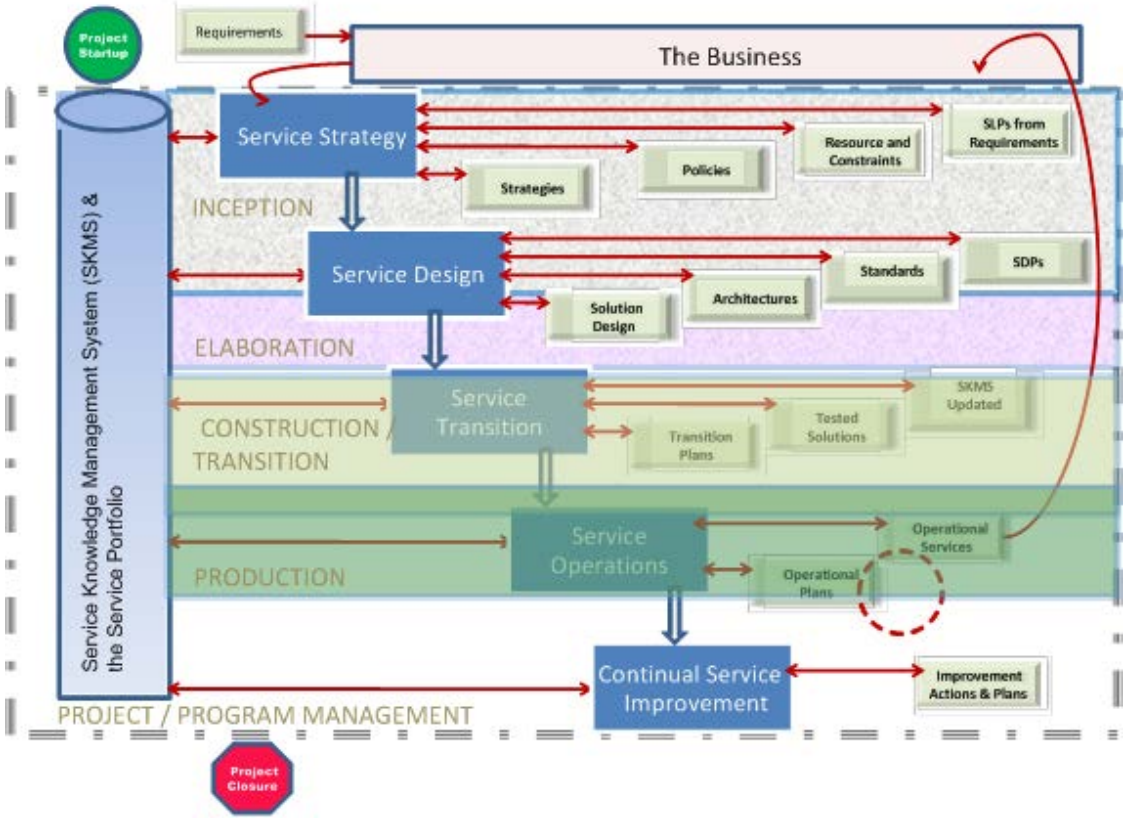
A service at any time can be 'retired,' especially when it is progressing as a 'project' in the pipeline phase. For example, during the recent 'Credit Crunch' it has been necessary for many organizations to review their current Service Portfolio and in some cases 'shelve,' or in other words, 'retire' services. In the future, these services may well be re-initiated and follow the Service Lifecycle again.

The following figure illustrates where the various Service Status values fit within the ITIL Service Lifecycle.



Service Portfolio and Lifecycle

The following figure shows the synergy between OUM and ITIL, illustrating how the various ITIL processes map to the OUM Implement focus area project lifecycle. In this example, the service lifecycle is initiated by a change in business requirements.



OUM and ITIL Synergy

To highlight the similarities between OUM and ITIL in the diagram above, ITIL V3 Service Lifecycle phases are superimposed over the OUM Implement focus area phases to reflect how the respective phase components are related. The small green rectangular boxes illustrate some of the common activities. Both frameworks focus on providing value to the business and address managing quality, risk and accountability. Lifecycle phases within OUM and ITIL have defined entrance and exit criteria. Activities are tracked, measured, and evaluated against critical success factors. Corrective actions are taken as needed to ensure the business requirements are met. OUM and ITIL focus on repeatability and reuse of work components with an emphasis on identifying opportunities for improvement.

From the perspective of ITIL, requirements are identified and agreed upon within the Service Strategy phase. The requirements and the defined set of business outcomes are documented within a Service Level Package (SLP). The SLP is passed to the design phase In the design phase is where a service solution is produced along with a Service Design Package (SDP) containing everything necessary to take the service through the remaining stages of the lifecycle. The SDP is passed to the Service transition phase where the service is evaluated, tested and validated. The Service Knowledge Management System (SKMS) is updated and the service is transitioned into the live environment where it enters the Service Operations phase. The Service Knowledge Management System is the central repository of the IT organization's data, information and knowledge. The SKMS consists of a set of tools and databases that are used to manage knowledge and document the IT ecosystem. Whenever possible, the ITIL Continual Service Improvement Process identifies opportunities for the improvement of weaknesses or failures anywhere within any of the lifecycle phases.

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ITIL TO OUM MAPPING

The following tables compare the Services Management lifecycle phases and processes described within ITIL with the Implement and Manage focus area lifecycle phases and processes described within OUM.

ITIL defines a process as a structured set of activities designed to accomplish a specific objective. A process takes one or more defined inputs and turns them into defined outputs. A process may include roles, responsibilities, tools and management controls required to reliably deliver the outputs. A process may define policies, standards, guidelines, activities, and work instructions required to carry out the process. A process is considered a closed-loop system because the results of the process typically provide for change or transformation of an event to a specific goal. Feedback about the results of a process provides improvement opportunities.

An ITIL function is defined as a team or group of people specialized to perform certain types of work and responsible for specific outcomes. A function is self contained with capabilities and resources such as work methods, knowledge and experience. Coordination between functions is achieved by those groups having responsibilities within shared processes. An example of a function is the Service Desk. A Service Desk is staffed by a group of people who provide a central point of contact for the reporting of difficulties, complaints or questions and are critical to the operation of the Incident Management process.

ITIL Lifecycle Phases and Processes Mapped to OUM Implement and Manage Focus Areas Project Lifecycle Phases

Continual Service Improvement is responsible for managing feedback from each process throughout the service lifecycle. The feedback is used to improve programs and projects based on strategic objectives.

ITIL LIFECYCLE PHASE	ITIL PROCESS / FUNCTION NAMES	TYPE		OUM IMPLEMENT AND MANAGE FOCUS AREAS PROJECT LIFECYCLE PHASES						
		ITIL Process	ITIL Function	Project Start Up	Inception	Elaboration	Construction	Transition	Production	Project Closure
Service Strategy	Service Portfolio Management	X								
	Demand Management	X								
	IT Financial Management	X								
Service Design	Service Level Management	X								
	Service Catalog Management	X								
	Availability Management	X								
	Capacity Management	X								
	Supplier Management	X								
	Information Security Management	X								
	IT Continuity Management	X								
Service Transition	Change Management	X								
	Service Asset and Configuration Management	X								
	Release and Deployment Management	X								
	Knowledge Management	X								
	Support and Transition Management	X								
	Service Validation and Testing	X								
	Service Evaluation	X								
Service Operation	Event Management	X								
	Incident Management	X								
	Problem Management	X								
	Access Management	X								
	Request Fulfillment	X								
	Service Desk		X							
	Technical Management		X							
	IT Operations Management		X							
	Application Management		X							
Continual Service Improvement	Service Measurement	X								
	Service Reporting	X								
	7 Step Improvement	X								

OUM Implement and Manage Focus Area Lifecycle Phases and Processes Mapped to ITIL Lifecycle Phases

The OUM Manage Project Execution and Control phase includes tracking the project's accomplishments and reviewing results against documented WBS-dictionary, project estimates, time schedule, resources plan and cost budget. Plans are adjusted based on the actual accomplishment and results.

OUM IMPLEMENT / MANAGE LIFECYLE PHASE	OUM PROCESS NAME	ITIL SERVICE LIFECYCLE PHASES			
		CONTINUAL SERVICE IMPROVEMENT			
		STRATEGY	DESIGN	TRANSITION	OPERATION
Project Start Up	Bid Transition				
Inception	Business Requirements				
	Requirements Analysis				
	Design				
	Implementation				
	Testing				
	Performance Management				
	Technical Architecture				
	Data Acquisition and Conversion				
	Documentation				
	Organizational Change Management				
	Training				
Elaboration	Business Requirements				
	Requirements Analysis				
	Analysis				
	Design				
	Implementation				
	Testing				
	Performance Management				
	Technical Architecture				
	Data Acquisition and Conversion				
	Documentation				
	Organizational Change Management				
	Training				
Construction	Business Requirements				
	Requirements Analysis				
	Analysis				
	Design				
	Implementation				
	Testing				
	Performance Management				
	Technical Architecture				
	Documentation				
	Data Acquisition and Conversion				
	Organizational Change Management				
	Training				
	Transition				
Transition	Testing				

Transition	Testing				
	Performance Management				
	Data Acquisition and Conversion				
	Documentation				
	Organizational Change Management				
	Training				
	Transition				
Production	Performance Management				
	Organizational Change Management				
	Operations and Support				
Project Closure	Gain Acceptance				
	Close Processes and Contracts				
	Document Lessons Learned and Archive Project				
Program Management	Execute and Control Program Management				

ITIL addresses how IT organizations as a whole should operate; the OUM Implement and Manage focus areas, which are the center point of the mapping above, address how Oracle solution implementation projects should be executed. Both frameworks rely heavily on process and the use of tools to enable consistent execution of processes. The ITIL framework and the OUM framework support each other in a way that propels services and operations to a greater level of proficiency. Both frameworks address the need to manage quality, risk, and accountability.

While the tables above serve to illustrate the full extent of the common areas shared by OUM and ITIL, the supplemental guidelines provided in this version of the guide focus on the ITIL best practices contributing to the lifecycle management of the My Oracle Support Services portal, which are concentrated in the Service Design and Service Transition lifecycle phases of the ITIL framework.

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ORACLE SUPPORT SERVICES SUPPLEMENTAL TASK GUIDELINES FOR LIFECYCLE MANAGEMENT OF THE MY ORACLE SUPPORT SERVICES PORTAL

This component contains OUM supplemental task guidance focused on best practices contributing to the lifecycle management of the My Oracle Support Services Portal. The guidance complements and extends the baseline OUM guidance and should be used in conjunction with the guidelines in the OUM Task Overviews.

- The first column of the following table provides a complete list of all tasks in this document for which "supplemental" guidelines are provided as well as a quick link to the "supplemental" task guideline in this document.
- The second column provides a link to the OUM Task Overview.

The task guidelines in this document should be used to supplement the guidelines that appear in the OUM Task Overview. The guidance included in this component is focused on best practices contributing to the lifecycle management of the My Oracle Support Services portal.

Oracle Support Services - OUM Supplemental Task Guidelines (this document) collapse table | expand table	OUM Task Overview
Technical Architecture	
TA.060 Define System Operations and Management Strategy	TA.060 Define System Operations and Management and Strategy
TA.090 Develop Security and Control Strategy	TA.090 Develop Security and Control Strategy
TA.100 Define System Management Procedures	TA.100 Define System Management Procedures
Transition	
TS.040 Design Production Support Infrastructure	TS.040 Design Production Support Infrastructure
TS.052 Implement Production Support Infrastructure	TS.052 Implement Production Support Infrastructure

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TECHNICAL ARCHITECTURE (TA) PROCESS GUIDANCE

TA.060 Define System Operations and Management Strategy

My Oracle Support access and use should be a key component of the System Operations and Management Strategy. Its proper employment, in conjunction with other operations management tools and procedures, can contribute significantly to the establishment of a personalized, proactive and collaborative environment for managing the system. From an ITIL perspective, the definition of the System Operations and Management Strategy is typically done within the ITIL Service Design Lifecycle phase. To ensure effective use of My Oracle Support the System Operations and Management Strategy should address the following recommended best practices contributing to the lifecycle management of the My Oracle Support Services portal:

- **Establish Policies for My Oracle Support Browser Usage** - Establish process and procedures for managing the My Oracle Support browser requirements. Plan the installation, configuration and maintenance of the browsers. Consider alternative solutions for use of browser functionality within the oracle.com domain. Refer to the Establish Browser Policies section for guidance on establishing browser policies.
- **Define Oracle Configuration Manager Configuration** - Implement Oracle Configuration Manager to centralize configuration information. Oracle Configuration Manager tracks key Oracle system details, providing essential data to manage the Oracle system environment. Oracle Configuration Manager is used to personalize the support experience by collecting configuration information and uploading it to the Oracle repository. When customer configuration data is uploaded on a regular basis, customer support representatives can analyze this data and provide better service to the customer. For example, when a customer logs a service request, the configuration data can be associated directly with that service request. The customer support representative can then view the list of systems associated with the customer and solve problems accordingly. The Oracle Configuration Manager Installation and Administration Guide provides configuration guidance.
- **View Oracle Configuration Manager Best Practices** - View Best Practices to understand how Oracle Configuration Manager can be used to maintain system health. The Oracle Configuration Manager Quick Start Guide contains best practices for the lifecycle management of Oracle Configuration Manager.

TA.090 Develop Security and Control Strategy

When implementing any new system involving the installation of software, there is always a concern that line of business security may be compromised. My Oracle Support access and use should be a key component of the Security and Control Strategy. From an ITIL perspective, development of the Security and Control Strategy would typically be done within the ITIL Service Design Lifecycle phase. To ensure the information assets of the business are secure, the Security and Control Strategy should address the following recommended best practices contributing to the lifecycle management of the My Oracle Support Services portal:

- **Perform My Oracle Support Security Access Control Configuration** - Define security requirements to secure user access to My Oracle Support based on users' roles and responsibilities. Assess security risks and establish controls to mitigate identified risks. Establish policies, processes and procedures for managing My Oracle Support security controls to include data security and functional access. Establish plans for monitoring, reporting, and auditing for compliance.
- **View My Oracle Support Security Access Control Best Practices** - View best practices to understand the My Oracle Support access control capabilities and how to use them to maintain security and segregation of duties.
- **View My Oracle Support Roles and Privileges Best Practices** - View best practices to understand the My Oracle Support user roles and the responsibilities of the roles.
- **Complete Initial My Oracle Support Account Registration** - Complete user registration for getting access to the My Oracle Support portal.
- **Log In To My Oracle Support to Verify Registration** - Log in to the My Oracle Support Portal to verify registration.

TA.100 Develop System Management Procedures

System Management is the process by which the system is controlled, operated, maintained and monitored with allowances for additions and / or changes as necessary to meet business requirements. System Management procedures should be documented and available to ensure that the system is operated, maintained and modified in a controlled manner. My Oracle Support should be a key component of the System Management Procedures. From an ITIL perspective, defining requirements for System Management Procedures would typically be done within the ITIL Service Design Lifecycle phase. To ensure effective use of My Oracle Support, the System Management Procedures should address the following recommended best practices contributing to the lifecycle management of an Oracle investment.

- **Define Customer User Administrator Processes** - Manage the My Oracle Support access control processes. The Oracle Customer User Administrator is responsible for securing the access and use of the data and services available through My Oracle Support. Establish management procedures for administering My Oracle Support and user system access. Establish plans for monitoring, reporting, and auditing for compliance.
- **View Customer User Administrator Best Practices** - View best practices to understand the responsibilities of the customer user administrator. Understand how to use My Oracle Support functionality to maintain My Oracle Support security access control.
- **Identify Customer User Administrators** – Define a System Management Procedure to ensure a minimum of two customer user administrators are assigned to manage user access to My Oracle Support. An overview of the roles and responsibilities of the Customer User Administrator can be found here.
- **View Working Effectively with Support Best Practices** - View Working Effectively With Global Customer Support [ID 166650.1] best practices to understand how to effectively collaborate with Oracle support.
- **View My Oracle Support Reporting Best Practices** - View best practices to understand how to use My Oracle Support reporting capabilities.
- **Establish Training for My Oracle Support Services** - Establish training to understand how to use the capabilities of My Oracle Support services to manage Oracle investments. Oracle has recorded training sessions that will provide beneficial tips and information to help with specific situations.

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TRANSITION (TS) PROCESS GUIDANCE

TS.040 Design Production Support Infrastructure

The Production Support Infrastructure involves the management and coordination of the implementation of the processes, and systems, which from an ITIL perspective, are documented during the ITIL Service Design Lifecycle phase. My Oracle Support should be a key component of the Design Production Support Infrastructure. In designing your production support infrastructure, ensure that you provide for a test environment that is a logical match for the production environment to allow reproduction and resolution of incidents. The Design Production Support Infrastructure task description provides guidance for the development and improvement of capabilities for transitioning the production support infrastructure into operations. Design Production Support infrastructure would typically be done within the ITIL Service Transition Lifecycle phase. To ensure effective transition to production operations, Design Production Support Infrastructure should address the following recommended best practices contributing to the lifecycle management of your Oracle investment:

- **Perform My Oracle Support Best Practice Recommendations** - Establish awareness of My Oracle Support best practice recommendations. Include My Oracle Support best practice recommendations in the design of your production support infrastructure. The total cost of ownership of the Oracle investment is improved with the adoption of My Oracle Support best practices.
- **Establish Policies for My Oracle Support Health Check Usage** - Apply best practice guidance to establish policies for using My Oracle Support health checks. Use of My Oracle Support health checks improves availability and minimizes the impact of service disruptions.
- **Establish Policies for My Oracle Support Patch Plans** - Apply best practice guidance to establish policies for using My Oracle Support patch plans. Use of My Oracle Support patch plans improves availability and minimizes the impact of service disruptions.
- **View My Oracle Support Projects Best Practices** - View best practices to understand My Oracle Support projects functionality and how My Oracle Support projects can be used to manage system health.
- **View Oracle Web Conferencing Best Practices** - View best practices to understand how to use Oracle Web Conferencing to prepare for online collaboration with Oracle support.
- **View Oracle Communities Best Practices** - View best practices to understand how to collaborate with Oracle communities. Oracle communities provide a valuable avenue for customers to share knowledge.

TS.052 Implement Production Support Infrastructure

The Implementation of the Production Support Infrastructure involves the building, testing and deployment of the Design Production Support Infrastructure. My Oracle Support should be a key component of the Implement Production Support Infrastructure. The Implement Production Support Infrastructure task description provides guidance to put into effect efficient repeatable build and test mechanisms to ensure effective transition to production. From an ITIL perspective, Implement Production Support infrastructure would typically be done within the ITIL Service Transition Lifecycle phase. Implement Production Support Infrastructure should address the following

recommended best practices contributing to the lifecycle management of the My Oracle Support Services portal:

- **Implement Support Test Environment** – Implement the design of your production support infrastructure. Validate the implemented test environment is a logical match for the production environment.
- **Verify External Internet Connectivity** - All support services tasks depend upon having external internet access. Test My Oracle Support Portal to determine connectivity to external sources.
- **Install Adobe PDF Requirements for My Oracle Support** – Install Adobe PDF requirements on end user desktops.
- **Configure Adobe PDF Requirements for My Oracle Support** - Configure Adobe PDF settings on end user desktops.
- **Verify My Oracle Support Browser Desktop Installation** - Confirm supported My Oracle Support browsers are installed, configured and available for use on end user desktops.
- **Configure Oracle Web Conferencing** - Configure Oracle Web Conferencing settings on end user desktops to prepare for online collaboration with Oracle Support
- **Verify Oracle Web Conferencing Compatibility** - Run the Oracle Web Conferencing compatibility test to confirm desktop browsers are configured properly and ready for collaboration with Oracle Support.
- **Configure Customer User Administrator My Oracle Support Account** - Register administrators with My Oracle Support
- **Administer Customer Support Identifiers** - Administer Customer Support Identifiers to ensure users have access to the currently licensed products within Oracle support services. Access to My Oracle Support is controlled through user assignments to Customer Support Identifiers.
- **Configure Default Customer Support Identifier Privileges** - Configure the Default Privileges for the Customer Support Identifiers administered by the customer user administrator.
- **Administer My Oracle Support User Security Access** - Administer Role-Based Access to My Oracle Support functionality in accordance to corporate security policy.
- **View My Oracle Support Personalization Best Practices** - View best practices to understand the personalization capabilities of My Oracle Support.
- **Configure My Oracle Support Service Request Profiles** - Configure the My Oracle Support service request profiles. The service request profiles serve as templates to streamline the create service request process. To configure this setting, go to the settings tab located at the left hand column, and select Service Request profiles under Personal settings.
- **Configure Hot Topic Email Preferences** - Configure personalization settings to receive email messages to create awareness for new or updated topics of interest.
- **Configure My Oracle Support Dashboard Regions** - Use the customize button in the Dashboard to configure specific details through various customizable regions that include Service Requests. The Dashboard tab provides you with a concise, personalized, dynamic view of your systems and interactions with Oracle Support.
- **Configure My Oracle Support PowerView Preferences** - Configure PowerView to take advantage of the My Oracle Support global search filtering capabilities. To configure this setting, go to the PowerView settings located at the top-left of the Dashboard.
- **Configure My Oracle Support Alert Preferences** - Subscribe to My Oracle Support alerts relevant to business responsibilities to keep current on information that can help mitigate service disruption. To configure this setting, go to the Knowledge Articles region and select the modify settings located at the top-right corner of the region.
- **Configure My Oracle Support Newsletter Preferences** - Subscribe to Oracle support newsletters that are relevant to the business to keep current on information. Oracle support teams provide news and technical updates through newsletters.
- **Install Oracle Configuration Manager Per Established Policies** - Install Configuration Manager per established policies for use with My Oracle Support if not already done so.
- **Configure Oracle Configuration Manager** - Configure Oracle Configuration Manager to comply with business requirements.
- **Schedule Oracle Configuration Manager Collections** - Schedule the collection of configuration data to track Oracle environment configuration details.
- **Execute Trial Run of Oracle Configuration** - Execute a trial run of Oracle Configuration Manager to verify everything works as expected.
- **Configure My Oracle Support Health Checks** - Configure My Oracle Support health check settings to align with business requirements. To configure this setting, go to the Dashboard and modify settings in the Health Recommendations region.
- **Configure My Oracle Support Patch Recommendations** - Configure My Oracle Support patch recommendations to align with business requirements. To configure this setting, go to the Dashboard and modify settings in the Patch Recommendations region.
- **Configure My Oracle Support Projects** - Configure projects to get assistance in managing complex support scenarios. Projects provide Oracle support a contextual view of the business. To configure this setting, go to the Dashboard and modify settings in the Projects region.
- **Subscribe to My Oracle Support Communities** - Subscribe to communities within the Oracle domain that are of interest to enable instant connection to a network of peers, industry experts, and Oracle product specialists.

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ORACLE SUPPORT SERVICES SUPPLEMENTAL TASK GUIDELINES FOR IT CHANGE MANAGEMENT

This component contains OUM supplemental task guidance for managing IT changes in an Oracle environment. The guidance complements and extends the baseline OUM guidance and should be used in conjunction with the guidelines in the OUM Task Overviews.

- The first column of the following table provides a complete list of all tasks in this document for which "supplemental" guidelines are provided as well as a quick link to the "supplemental" task guideline in this document.
- The second column provides a link to the OUM Task Overview.

The task guidelines in this document should be used to supplement the guidelines that appear in the OUM Task Overview. The guidance included in this component is focused on on Oracle's recommended approach and best practices for handling IT changes to an Oracle Eco System. Implementing the recommended technical solutions that consist of process, governance, product capabilities and knowledge ensures that standardized methods and procedures are used for efficient and prompt detection and handling of all changes.

Oracle Support Services - OUM Supplemental Task Guidelines (this document) collapse table | expand table	OUM Task Overview
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RKM.050 Monitor and Control Risk	RKM.050 Monitor and Control Risk
Quality Management	
QM.020 Develop and Document Quality Control and Reporting Process	QM.020 Develop and Document Quality Control and Reporting Process
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CM.010 Develop Configuration Management Strategy and Processes	CM.010 Develop Configuration Management Strategy and Processes
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RISK MANAGEMENT (RKM) PROCESS GUIDANCE

Risk is a key input into Change Management, and also a key output. IT environments should be monitored for risks, and risks identified, should be analyzed and mitigated in a timely manner to ensure system health and security. To ensure risks due to changes to the IT are identified and mitigated, the following Risk Management tasks should address the recommended best practices contributing to IT Change Management, as outlined below. (See supplemental guidance for task TA.020, Define Technical Architecture Requirements and Strategy for a list of changes.)

RKM.040 Manage Risk

Changes are typically analyzed for potential risks to the system. Changes are also proactively identified as described in the task, Monitor and Control Risk (RKM.050), and analyzed for risks. Each identified risk is evaluated and mitigated accordingly. To effectively manage changes, the Manage Risk task should address the following recommended best practices:

- **Analyze Changes** - Once a change to the IT environment is logged, both the Business and Technical impact are analyzed. This helps manage the risk of the changes introduced. The change may not just impact that system, but also inter-connected systems. For example: An eCommerce application may make a call-out to a payment processing application like "VISA". Capturing associations and relationships between configuration items, services and systems allows for easy "what is affected, what is impacted" analysis. Oracle Enterprise Manager EM12C can be utilized to traverse the relationships from the bottom to the top of the tree to see if a problem will occur if changes are made to the element. It answers the question: What items are dependent on my element that would be affected should I do something to my element. For example, if I shut down a listener, what databases would be affected?

RKM.050 Monitor and Control Risk

Risks need to be proactively monitored and controlled so that the IT system is healthy. The risks identified in the execution of this task are managed through execution of task RKM.040 (Manage Risk). Monitor and Control Risk should address the following best practices to effectively identify potential risks to the system and suggest appropriate changes to mitigate the risk:

- **Identify Changes to Monitor** – There are several sources or inputs for change. Some common sources are listed below but your business requirements may include additional sources of change to monitor.

- **Patches and Security Updates:** Patch Recommendations(changes) to remain stable, supportable and current, provided through Advisories in OEM/MOS.
- **Unauthorized Changes and System Performance:** Automatic Real time Change Detection through healthchecks, risk analysis, monitoring tools.
- **Industry and/or Regulatory Requirements:** Compliance to Industry/Government Standards.
- **Changes due to Business Requirements:** IT changes as a result of business requirements like business setups, deployments, enabling new features, etc.

- **Create Policies to Monitor for Changes** – Policies or rules are effective in managing configuration drift (through installation of patches, adding files and directories, changing settings and ports, editing its dependencies, and so forth) by continually auditing against prescribed configurations. Notification rules can be applied and corrective actions can be assigned for the defined policies. Oracle Enterprise Manager EM12C can be used to create policies/rules to evaluate the compliance of targets and systems as they relate to business best practices for configuration, security, and storage.

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QUALITY MANAGEMENT (QM) PROCESS GUIDANCE

Like any other process, change management needs to be measured and continually improved upon. The success criteria are defined and then the improvement plan is created at a later point. Metrics identified in the improvement plan are measured to create the action plan for improvement. To ensure that the Change Management process is monitored and continually improved, the following Quality Management tasks should address the recommended best practices contributing to IT Change Management, as outlined below:

QM.020 Develop and Document Quality Control and Reporting Process

Identifying Success factors and metrics are key to drive improvement. In this task we will document a Quality Control and Reporting process to measure the efficiency and effectiveness of IT Change Management process. The Quality Control and Reporting Process should address the following best practices to effectively identify Key Performance Indicators and Success Factors to continually drive improvements in the IT Change Management Process:

- **Identify Success Criteria** - The types of measures that need to be considered include:
  - **Output Measures:** Metrics that measure the outcome of the process or answer the question, Has it made a difference? An example is a reduction in the disruptions caused by unsuccessful changes.
  - **Workload Measures:** Metrics that measure total number of changes to the environment.
  - **Process Measures:** Metrics that measure performance, including perceptions and satisfaction levels with the overall process, e.g., speed, ease of use, etc.

    The maturity level of the process will drive the types of measurements appropriate to be collected. In a new process, more process measures and workload measures are appropriate, i.e., Do changes have back-out plans? Have they been tested? What is the breakdown of the type of changes? As the process matures, business-outcome / output measures are more appropriate, such as reduction in impact of changes.

- **Create Continual Improvement Plan** - As for any process, create a Continual Improvement Plan by measuring performance. Here are some sample metrics:

  Measure Process:

  1. % Rejected RFCs
  2. Unauthorized Changes Detected
  3. % of Change Requests (Business-Driven Need) Implemented On Time
  4. Average Time to Make Changes
  5. The Change Backlog
  6. % Fewer Changes 'Backed Out' Because of Testing Failures
  7. Changes Required by Previous Change Failures
  8. % of Reports Produced on Schedule

  Quick and Accurate Changes:

  1. The Number of Urgent Changes

2. Urgent Changes Causing Incidents
3. Changes Implemented Without Being Tested
4. Urgent Changes Requiring Back-Out
5. % of Urgent or High Priority Changes Submitted Without Business Case to Justify Decision

Protect Service:

1. Amount of Both the Scheduled and Unscheduled Service Unavailability Caused by Changes
2. Changes Backed Out
3. Unsuccessful Changes
4. Changes Causing Incidents
5. Changes Impacting Core Service Time and SLA Service Hours
6. Changes Activated Outside Core Service Time and SLA Service Hours
7. Changes not Referred to a Change Advisory Board (CAB) or CAB Emergency Committee
8. Customer Satisfaction Survey (CSS) Feedback on Change
9. Failed Changes that do not have Recorded Back-Out Plan
10. Time to Implement a Change Freeze

Efficiency and Effectiveness Results:

1. % Efficiency Based on Number of RFCs Processed
2. The Accuracy of Change Estimates
3. The Average Cost of Handling a Change
4. Change Overtime due to Better Planning
5. The 'Cost' of Failed Changes
6. % of Changes Implemented On Time
7. % of Changes Implemented to Budget
8. Failed Changes
9. Backed Out Changes

Tools for Measuring Metrics:

Configure your existing Change Management tools for data collection. In Oracle's solution, the reporting mechanisms in Oracle Enterprise Manager and Siebel Help Desk can be used to collect performance data and display it in a dashboard to drive continual improvement.

- **Review Reports to Drive Improvement** - The reports established in the implementation phase are recommended to be viewed on a regular basis to drive continuous improvement on the Change Management process.

  How to view reports?

  - **Change Identification and Deployment** - The generic Enterprise Manager Reports Web site URL is http://hostname:port/em/public/reports. The URL for this site is automatically set when Enterprise Manager is installed. A link to this site can be found on the Create/Edit Report Definition Access page and the Reports tab.
  - **Change Tracking** - The Change tracking tool typically has reports to manage performance of changes such as number of emergency changes, failed changes which can be reviewed to drive improvement.

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CONFIGURATION MANAGEMENT (CM) PROCESS GUIDANCE

Configuration Management is a prerequisite for all the other IT processes. Configuration management is intrinsically linked to change management to ensure that a clear

and complete "snapshot" of the environment is always available. For effective change management, assets are discovered and maintained in a configuration database (widely known as CMDB). From this point on, changes to and status of the configuration items are recorded in the configuration management database. To ensure effective rollout of changes on configurations, the following Configuration Management tasks should address the recommended best practices contributing to IT Change Management, as outlined below:

CM.010 Develop Configuration Management Strategy and Processes

Change Management is part of the lifecycle of a configuration item. Understanding the Change Management Process is a key factor that facilitates developing a Configuration Management Strategy and Process.

- **Understand/Review Change Management Process** – IT Change Management is the process of identifying, requesting, analyzing, approving, developing, implementing, and reviewing planned or unplanned changes to assets/configurations within the IT infrastructure. The Change Management Process begins with the identification of a Change Request and ends with the satisfactory implementation of the change and the communication of the result of that change to all interested parties. Change Management process consists of the following steps:
    - **Identify/Detect a Change** - Determine when a change needs to be made or when unauthorized changes have been made proactively before it negatively impacts the business.
    - **Formally Request a Change** - Document all change requests by creating a Request for Change (RFC).
    - **Categorize and Prioritize the Change** - Prioritize the change after assessing the urgency and the impact of the change on the infrastructure, end user productivity and budget.
    - **Analyze and Justify the Change** - Develop specific justification for the change by analyzing the impact on the infrastructure, business operations, and budget.Use this information to develop an extensive risk and impact analysis.
    - **Approve and Schedule the Change** - Review the risk/impact analysis to approve the change by routing the Request for Change (RFC) to technical and business approvers as appropriate. In the event of a major or significant change, ensure that committees such as Change Advisory Board (CAB) are involved for approval or rejection of the change.
    - **Plan and Deploy the Change** - Develop and deploy the change reviewing the specific implementation steps in a manner that will minimize impact on the infrastructure and end users.
    - Conduct Post-Implementation Review - Conduct a post-implementation review to ensure whether the change has achieved the desired goals. Post-implementation actions include deciding to accept, modify or back-out the change; contacting the end user to validate success; and finalizing the change documentation.

CM.020 Obtain Configuration Management Tools

To effectively manage Oracle IT Assets, you need to identify the Tools which will help you with the process. This section provides supplemental guidance on identifying the Tools and Services for effective Configuration and Change Management of the Oracle infrastructure.

- **Identify Tools and Services** - Change management tools and services that help you facilitate a successful implementation and stay ahead of constant change. With well defined processes and tools available to help plan, execute, and track changes you can take a proactive stance to manage changes in your Oracle environment.
    - **Products and Tools:** Tools can help reduce risk, minimize impact, avoid business disruption, and most importantly, increase your first-time success rate when preparing for and managing change in your infrastructure.
      Below is a list of Change Management Products/Tools provided by Oracle:

Process	**Product / Tool**
Identify and Deploy Changes	Oracle Enterprise Manager 12C
Integrated Support	Oracle Proactive Support Tools

Track and Manage Changes	Siebel Help Desk
	Oracle TeleService
	PeopleSoft Enterprise HelpDesk

- **Services:** Oracle's complete portfolio of services covers the entire solution lifecycle and has been developed to assist customers in reducing total cost of ownership, lowering risk and improving business value.
  Services related to Change Management are:

  1. My Oracle Support Patches and Updates Services
  2. Lifecycle Advisors
  3. ACS Services

  You can find additional services offered by Oracle Support in the Services Catalog.

CM.050 Create and Execute Software Release Management Plan

Changes identified as part of the Change Management process described in the task, Create and Execute Configuration Control Plan (CM.070) are released using specific release plans. This section provides best practices to strategize these releases using maintenance plans and promoting the changes through different environments (such as, DEV, TEST, PROD) and applying the changes to PROD environment.

- **Strategize Release of Changes** - Changes can be either planned (proactive) or emergency (reactive). Plan roll out (release) of proactive changes ahead of time using pre-defined maintenance schedules and allocate regular maintenance windows for reactive (emergency) changes. The Change Manager in addition to business unit representatives and members of the Change Advisory Board (CAB) will establish maintenance windows in alignment with the business cycle in order to maintain system stability and business continuity. Changes should be frozen during peak business processing times, e.g. End of Month, End of Quarter, End of Year dependent on the business processing cycle.

  Example - Steps to Create a Maintenance Plan
  - Define Change Freeze Periods
    - Fiscal Year(FY) June1 - May 31
    - First Quarter(1Q) - June 1 - Aug 31
    - Second Quarter(2Q) - September1 - November 30
    - Third Quarter(3Q) - December1-February 28
    - Fourth Quarter(4Q) - March1-May31
    - Hard freeze is the time period when quarterly closes are done (just before each quarter ends) when there are no changes permitted at all.
    - Soft Freeze is the time period when there are still some quarterly close activities taking place, but emergency changes are permitted. Major changes are not permitted during the Soft Freeze.

    Quarterly Freeze Periods

Freeze	**1QFY12**	**2QFY12**	**3QFY12**	**4QFY12**
Hard Freeze	Aug18-Sep4	Nov15-Dec1	Feb14-Feb28	May14-Jun1
Soft Freeze	Sep5-Sep19	Dec2-Dec15	Mar1-Mar15	Jun2-Jun16

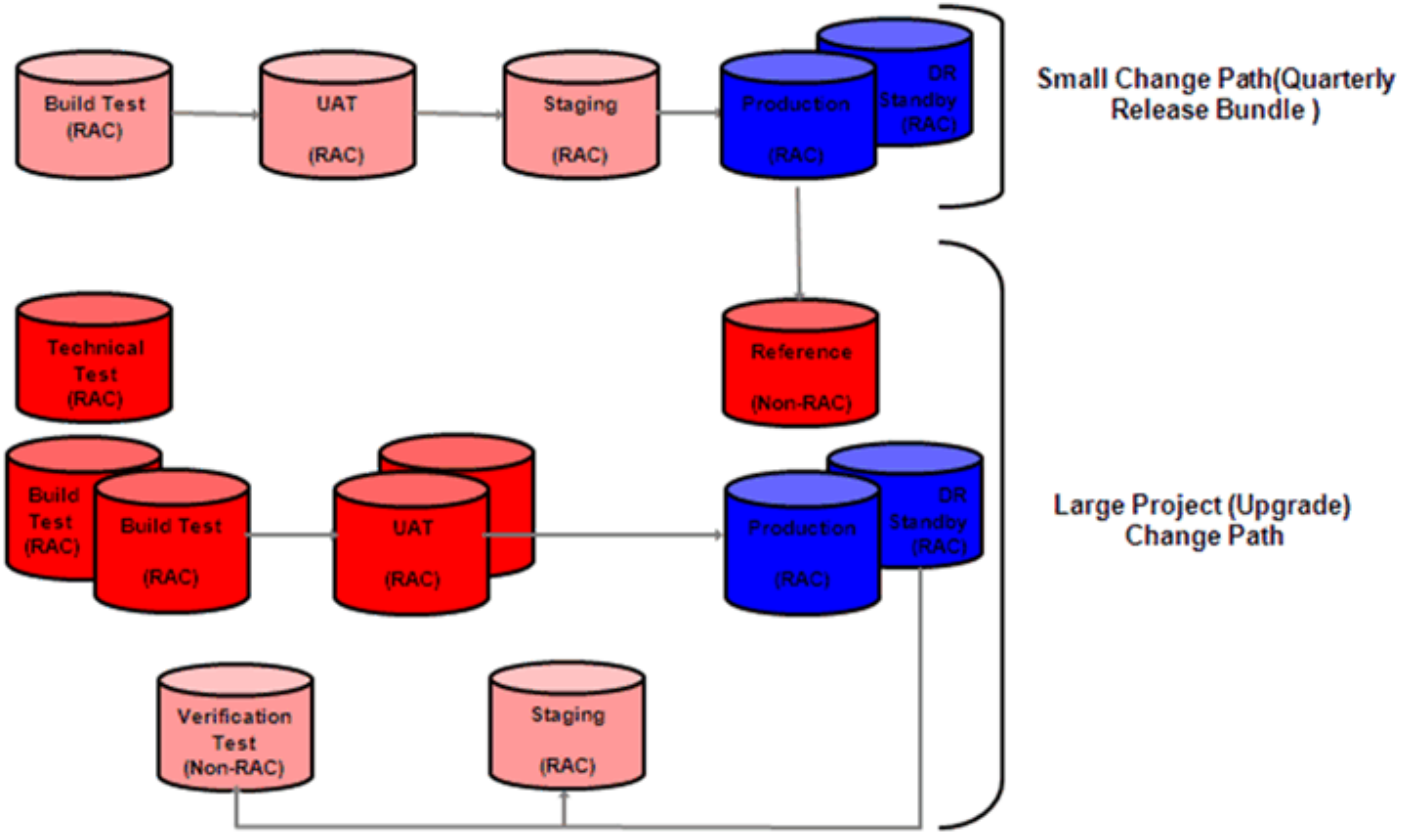
Create a Maintenance Plan - From the freeze periods, identify the available maintenance windows. Some examples of maintenance activities and their schedules follow:

- Operations Quarterly Maintenance would have simple configuration/performance changes, etc.
- Security patches are applied on a quarterly basis.
- Other types of changes could be scheduled as it fits the organization.

Quarterly Maintenance:

Freeze	1QFY12	2QFY12	3QFY12	4QFY12
Hard Freeze	Jul 24	Oct 23	Jan22	Apr 22
Soft Freeze	Jul 31	Oct 30	Jan 29	Apr 29

- **Implement Promotion of Changes** – One of the key steps in Release Management is progression of changes through various environments. Even though this depends on each organization, below are a couple of examples to promote changes:



- The first example shows a simple Change path which could be used for small-to-medium project changes(Example - Quarterly Release Bundle).
- The second example, shows how the promotion of changes might look like for larger projects (like Upgrades).The Technical Test environment is a one-off test environment and has no designated change flow to other test environments. It is used for initial upgrade testing and performance testing. A copy of production is created just before go-live as a reference for issues which may arise. Upgrade changes will be introduced into normal production support path via refresh after project go-live.

- **Log and Manage Changes** – Part of the Software Release Management Plan and any changes released will consist of the following steps:
  - **Initiate Change:** A request for change (RFC) is initiated by the "Change Initiator" and here are some examples of how changes can be initiated and logged:

    1. Changed from an existing incident, where the solution can be to apply a patch, make a configuration change, etc. In this case the incident can initiate a Change Request using the ticketing system. Here is an example of doing it using Siebel Help Desk.
    2. For bigger changes (like an Upgrade roll-out), this is typically done by creating a Project in the ticketing system . Here is an example of doing this using Siebel Help Desk. You can also use My Oracle Support Projects to collaborate with Oracle Support on roll-out of major changes like an Upgrade.
    3. Changes can also be triggered by monitoring tools (like OEM) which can open a change ticket in the ticketing system (like Siebel) based on the criteria set using connectors as described in the Plan phase. Example - If there is a security alert raised as a result of non-compliance to the policies set in OEM, a change ticket can be initiated from OEM using the connector to populate the change record with the configuration information.
  - **Classify Change:** The classification of a change aims at providing a control factor for the change, which will be used for factual decision making with respect to change scheduling. Here is an example of how this can be done.
  - **Conduct Impact Analysis and Assess Risk:** Once a change is logged, both the Business and Technical impact are analyzed. The change may not just impact that system, but also inter-connected systems. For example: An eCommerce application may make a call-out to a payment processing application like "VISA". Capture associations and relationships between configuration items, services and systems allows for easy "what is affected, what is impacted" analysis. Here is an example of impact analysis on an EBS environment for rolling out patches:

    1. Apply the patch on a test environment that is a very recent clone or copy of your Production environment.
    2. Use Patch Wizard to assess technical impact (what files have been changed, etc.).
    3. Use Patch Wizard to assess Impact on Customizations based on the "Flagged files."
    4. The reading of the README file that comes bundled with the patch to learn what is new, what has changed, etc, especially for Release Update Packs.
    5. Oracle Communities has some forums for patch reviews where you can get information from other customers about patches they have applied and their results.
    6. In reactive or emergency patching situations where testing is not feasible, the patch may be applied on your Production system by using the AutoPatch (adpatch) test mode argument (apply=no).
- **Approve and Schedule Change** – Approval of a change is driven by its category which would have been defined when planning the architecture. Based on the Approval criteria setup during the Implement phase (CAB,ECAB, etc.), changes are authorized and scheduled using the Change Management system (Siebel Help Desk in our case). Here is an example of how this can be done.
- **Promote and Release Changes** – As part of Release Management, once the changes are logged, they are promoted through various environments and finally deployed in the Production system. Once they are deployed, changes are validated to ensure they have not introduced any regression and satisfy the requirement of implementing the change.
  - **Deploy Change** - Depending on the type of changes, simple changes are easy to deploy and can be done by instrumented change models, whereas major changes go through the entire change lifecycle of Development, Test and Production. Some examples of promotion of changes were covered in the "Implement phase".
    Below are some examples of how changes can be deployed using Oracle Solution:

    1. Product Upgrades - Use the Upgrade Advisors to guide you through an upgrade lifecycle between specific versions for a product or suite.
    2. Patches - Use the Patch Advisors to guide you through deployment of a patch.
    3. Changes in Application code and schema changes using OEM:
       - Automatically create new versions for new sets of planned changes.
       - Validate planned changes to identify conflicts or previous changes.
       - Preview and edit validated changes before applying.
       - Generate SQL script of final set of validated changes.
       - Apply validated planned changes.
  - **Release and Validate Change** - Once the changes have been tested and gotten the approval to be released, based on the maintenance schedule developed during the strategy phase, changes should be packaged and released.
    Some of the steps to be taken into consideration are:

1. GO-NOGO decision has been made
2. All the approvals are in place - should be tracked using the ticketing system (Siebel Help Desk)
3. Change is documented and communicated including the Projected Service Outage if there is any outage during the change release - through the ticketing system(Siebel Help Desk)
4. Transfer information to Operations and Support - create technical documents and training information as required. Oracle's User Productivity Kit (UPK) can be used to create training content.
5. Stabilize in the Production environment - Observe using System Management tool like OEM
6. Validate both from technical and business standpoint whether the change was successful and any lessons learned to drive improvements - can be done using feedbacks, technical comparison (new and old environments), Business process monitoring, etc.

CM.070 Create and Execute Configuration Control Plan

The Configuration Control Plan controls the changes to configurations to the IT infrastructure. It is important to both detect risks to the environment to proactively prompt the changes and also detect any unauthorized changes to the configuration. The detected changes are rolled out using release plans described in the task, Create and Execute Software Release Management Plan (CM.050). This section provides best practices to both identify health/patch recommendations and also unauthorized risk to IT infrastructure:

- **Detect Changes** - Assets and configurations are required to be evaluated related to business best practices for configuration, security and storage. There needs to be controls in place to ensure that the configurations are maintained properly. There are various sources of changes. Below are some examples of identifying the need to change or detect unauthorized changes. Once the changes are identified, using the Change Management Solution (example - OEM), the Connectors in OEM can be used to log and manage tickets in the ticketing system (example - Siebel Help Desk).
  - **Changes as a result of Configuration Drift:** A drift in configuration can be identified using Oracle Enterprise Manager (OEM) 12C across the stack, across system lifecycles, Baseline & Gold Std and 1 to 1, 1 to Many scenarios. Save the comparisons with versions so that it can be compared across various timelines. Automated notifications could also be setup. Example using OEM and Application Management Packs:
    - Find if all the E-Biz Suite configurations in the data center are following a particular standard.
    - Find the drift between Development, Stage and Production lifecycles.
    - Detect if peers are indeed similarly configured.
    - Reconcile differences, if any.
    - Revert back to an older snapshot upon deviation.
  - **Changes as a Result of Health and Patch Recommendations:** IT infrastructure is required to be monitored for system health. The recommendations out of the monitored system can be implemented using the Change Management Solution.

    Example: The critical Patch Advisory in OEM using the My Oracle Support integration provides:

    1. Vulnerability Analysis on the Security Patches (CPUs)
    2. Comprehensive Analysis of Applicable Patches across Products, Versions, and Platforms
    3. Supports both Online and Offline Modes
    4. Automated Patching Process
    5. Advisory and Vulnerability Reports

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INFRASTRUCTURE MANAGEMENT (IFM) PROCESS GUIDANCE

IFM.040 Manage and Maintain Infrastructure

To effectively deploy Change Management processes within an organization, there are infrastructure requirements to ensure changes are efficiently handled. Change

Management tools identified in CM.020 (Obtain Configuration Management Tools) need to be installed and maintained as part of the IT infrastructure being managed. To ensure we have a the necessary infrastructure to support roll-out of changes, the following Infrastructure Management tasks should address the recommended best practices contributing to IT Change Management, as outlined below:

- **Create Environments to Promote Change** - Changes are promoted from the Development environment all the way to Production where it is available for the users. While we understand that the types of environments vary across organization, below is a sample of what environments are required to roll out changes.

Environment	Description	Usage
Development	Initial Development Environment - IT Only	Used for Ongoing Development
Build Test	First Integrated Environment	Business Process Owners Involved
Unit Test	Full Test Environment	All Users along with Automated Tests
Stage	Final Test before Production	Ensure all Regression Tests (RTs) and Critical Flows work.
PROD	Production	Production

- **Setup Support Framework** - My Oracle Support is Oracle's Support framework. The following are the recommended steps in your infrastructure to integrate Support with your System Management tools (such as EM):

  1. Identify if Collector is Installed.
  2. Download and Install Oracle Configuration Manager(OCM) Collector:
     - If you had chosen to deploy OCM without the harvester, download and install collector in each ORACLE HOME. If there are lot of environments, the manual installation of Oracle Configuration Manager in these environments can be tedious and time consuming. Mass Deployment can facilitate the deployment of Oracle Configuration Manager in these environments, as well as update and configure various systems from a single location.
     - If you had chosen to deploy using OEM Harvester, it is sufficient you install one OCM collector for OMS instance in Connected mode. Then setup a Harvester job to gather and upload Configuration information to Oracle.
     3. Enable systems not having internet access using Oracle Support Hub utility.
  4. Validate OCM agents are running.
  5. Schedule Upload of Configuration Data - OCM has a scheduler process that dictates when collections occur. This can be configured using the command $ORACLE_HOME/ccr/bin/emCCR set collection_interval.
  6. Validate OCM data uploaded to MOS is current by checking the timestamp of the configuration.
  7. Subscribe to Communities for sharing information, posting questions and answers, and providing suggestions about Oracle products, services, and related technologies.
- **Setup Change Detection and Deployment Framework** - Oracle Enterprise Manager(OEM) is Oracle's solution for Change Detection and Deployment. The following information will enable you to setup the infrastructure required to manage your Oracle Environment:

  1. Install OEM 12C.
  2. Deploy OEM 12C Agent.
  3. Setup various options within OEM by clicking the "Setup" button at the top of the console:
     - Setup Roles and Administrators.
     - Setup notifications and format of emails.
     - Configure MOS within OEM (under Patching Setup).
     - Enable Management Packs(as licensed - Lifecycle management pack is key for change management).
     - Configure Connector for change management ticketing system (Management Connectors for Siebel, Remedy, etc.)

  4. Setup and Configure Software Library and stage the software images which can be accessed by any target.

- **Setup Change Approval and Tracking Framework** - Changes need to be managed and in order to automate this process, there are infrastructure requirements to install and setup the tools which will provide these capabilities. Even though Oracle offers quite a number of solutions, we have taken Siebel Help Desk as an example:

1. Install Siebel Help Desk Software.
2. Activate Help Desk Business Process and personalize it as required based on the strategy developed in the earlier phase using the Siebel Workflow Guide.
3. Configure auto escalation time limits and procedures for unresolved tickets.
4. Add Approval List in Siebel Help Desk.
5. Setup Projects in Siebel Help Desk for tracking major changes like Upgrade.

- **Implement Reporting Framework** - Reports are used to present a view of enterprise monitoring information for business intelligence purposes, but can also serve an administrative role by monitoring the performance of a process.

  Create Reports for Change Identification and Deployment - Below is a sample of how a reporting framework could be setup to identify and deploy changes using OEM:

  Information Publisher, Enterprise Manager's powerful reporting framework, makes information about your managed environment available to audiences across your enterprise. Information Publisher reports present an intuitive interface to critical decision-making information stored in the Management Repository while ensuring the security of this information by taking advantage of Enterprise Manager's security and access control.

  - **Review Out of Box Reports** - Information Publisher comes with a comprehensive library of predefined report definitions, allowing you to generate fully formatted HTML reports presenting critical operations and business information without any additional configuration or setup.
  - **Create Custom Reports** - Although the predefined report definitions that come with Information Publisher cover the most common reporting needs, you may want to create specialized reports. To create custom reports:

    1. Choose whether to modify an existing report definition or start from scratch. If an existing report definition closely matches your needs, it is easy to customize it by using Create Like function.
    2. Specify name, category, and sub-category. Grid Control provides default categories and sub-categories that are used for out-of-box reports. However, you can categorize custom reports in any way you like.
    3. Specify any time-period and/or target parameters. The report viewer will be prompted for these parameters while viewing the report.
    4. Add reporting elements. Reporting elements are pre-defined content building blocks, that allow you to add a variety of information to your report. Some examples of reporting elements are charts, tables, and images.
    5. Customize the report layout. Once you have assembled the reporting elements, you can customize the layout of the report.
  - **Schedule Reports** - Enterprise manager allows you to view reports interactively and/or schedule generation of reports on a flexible schedule. For example, you might want to generate an "Inventory Snapshot" report of all of the servers in your environment every day at midnight.

  Create Reports for Change Tracking - The lifecycle of change records and maintained in the ticketing system. Below are guidelines for setting this up for one of Oracle's Solution - Siebel Help Desk:

  - Install and Configure Reporting Framework
  - Create Reports
  - Schedule Reports

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ORGANIZATIONAL CHANGE MANAGEMENT (OCHM) PROCESS GUIDANCE

OCHM.010 Understand Customer's Organizational Change Management Strategy

IT Change Management is part of the Organization’s wider Change Management process and should blend with the organizational strategy towards managing changes. People, Process and Technology are three key parts of IT Management. In this section you will get Best Practices related to organizational structure (People part) to support IT Change Management.

Define Governance, Roles and Responsibility - Complexity of today's business and IT environments makes efficient and effective change management simultaneously more important and more difficult to achieve. For effective Change Management, it is very important to define proper Roles, Responsibilities, committees and approval hierarchy within an organization to ensure changes are coordinated and adequately governed.

Setup Roles - While this depends on each organization, here are some suggested roles to drive effective Change Management:

- **Global Business Owner** - Person responsible for the specific Business flow (for example, Order to Cash)
- **Global IT Solution Owner** - Person responsible for creating the solution to support the business flow (tools, technologies, applications to support the business flow). Global Business Owner and Global IT Solution Owner work very closely throughout the Change Management Lifecycle.
- **Change Manager** - Person responsible for managing the activities of the change management process for the IT organization. This individual focuses on the process as a whole more than on any individual change. The Change Manager is ultimately responsible for the successful implementation of any change to the IT environment.
- **Change Initiator** - Person responsible for originating changes by submitting a well defined change request(RFC). Change initiators could be users, suppliers or IT staff - depending on the nature of the proposed change.

Setup Committees - Within an organization there can be certain groups, comprised of cross functional representatives, who meet on a regularly basis to deliver on their specific charters. Delegates are identified to represent each of their respective areas on various committees. Here is an example of typical committees that can be setup appropriate to your requirement:

- **Change Advisory Board (CAB)** - The CAB is a cross-functional group set up to evaluate change requests for business need, priority, cost/benefit, and potential impacts to other systems or processes. Typically the CAB will make recommendations for implementation, further analysis, deferment, or cancellation. The CAB meets on a regular basis (depending on the size of the organization and dynamic nature of changes) to discuss changes which are pending for approval.
- **CAB Emergency Committee (CAB/EC)** - This is a subset of the CAB that deals only with emergency changes. It is established to be able to meet on short notice to authorize or reject changes with emergency priority.
- **Compliance Forum** - The Compliance Forum coordinates across the organization to drive Compliance. Meets as required (usually monthly). Attended by representatives from each Line of Business(LOB).
- **Security Committee** - The group's primary focus is to ensure globally consistent practices among the functional LOB support teams while facilitating and implementing security processes across all environments. They collaborate with the Security team in reviewing, approving and proposing security and access guidelines and processes.

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TECHNICAL ARCHITECTURE (TA) PROCESS GUIDANCE

TA.020 Define Technical Architecture Requirements and Strategy

For effective Change Management, technical requirements should be defined and planned so that the solution will cater the needs of the organization. As Change Management consists of selection of different tools and services based on the business requirements as described in CM.020 (Obtain Configuration Management Tools) , this section will provide Best Practices on defining technical requirements for implementing Change Management Solution in an organization.

- **Understand Best Practices** - While it is understood that each organization have different requirements, following are some of the Best Practices to consider when defining the Architecture for Change Management:

  1. Plan changes during scheduled maintenance windows and minimize ad hoc/emergency changes.
  2. Facilitate the introduction of all types of change via simple, clear and effective procedures and tools across the various environments.
  3. Assess all changes for feasibility (technology and cost) and business impact (risk).
  4. Perform due diligence appropriate to the Change category and type:
     - Reduce the breadth of approval on minor or standard changes wherever possible.
     - Increase the rigor and approval level for more significant changes.

5. Acquire Project management support and co-ordination.
6. Have Zero tolerance for unauthorized change.
7. Prioritize change by type: e.g. innovative versus preventive, versus detective, versus corrective.
8. Segregate duties, e.g. manager vs. approver vs. tester.
9. Establish a single focal point for managing change.
10. Enforce Change windows, establish authorization procedure for exceptions
11. Avoid change approvals without a tested back-out plan.

- **Define Structure for Change Management** - An important aspect of defining the architecture for Change Management Solution is to document the business requirements for change such as the change drivers, types of changes need to manage, how the changes will be promoted and having a clear and concise communication plan. While this depends a lot on the specific organization and business, below are some guidelines:
    - **Define Change Drivers** - In order to build a strategy for change, it is important to identify where changes come from or in other words what are the drivers for changes. Most of the changes are recurring except Implementation and Upgrade changes which are not very frequent.
        - **Business Driven** - Changes which initiate from the Business.
            - Implementation - When you want to implement a new Business process(Example - Implementing Order to Cash Business flow in your current installation). This would involve installation, configuration, testing, customization, roll out, etc.
            - Process improvement - Customization or setup to satisfy business requirements.
            - Optimization changes are made periodically when there are performance issues impacting business.
        - **Vendor or IT Initiated** - Changes which are IT or vendor driven to maintain supportability and currency:
            - Patch - regular maintenance activities to resolve issues, stay compliant with security, supportability, etc.
            - Upgrade - When you are upgrading from one major release to another(Example - Upgrading from Database 10G to 11G).
            - Configuration changes are made by IT periodically as part of maintenance.
    - **Define Types of Change Requests** - Different types of change requests can be defined based on the impact and in terms of how a change should be handled. Here are some examples:
        - **Emergency** - Having an immediate, detrimental effect on business operations. Streamlined approval process, may or may not go through full testing based on risk analysis.
        - **Urgent** - High impact but able to be scheduled in a maintenance window.
        - **Normal** - Scheduled in a standard maintenance window.
        - **Standard** - Low impact, repetitive change that is essentially pre-approved. Standard Change template should be defined. Depending on defined policy may not need CAB approval.
    - **Define Promotion of Changes** - Once changes are created, they are required to be properly tested and promoted to a Production environment. Here are some recommendations of environments:
        - **Development Environment** - This is the environment where the changes are developed.
        - **Test Environment** - Once the changes are developed, it is promoted to a Test environment where the changes are tested for its functionality. Test environment needs to match the Production environment as close as possible.
        - **User Acceptance (UAT)Environment** - Once the changes are individually tested, they are promoted to a UAT environment where the changes are regression and performance tested. UAT environment should be a very recent copy of Production.
        - **Stage Environment** - This is where the changes are finally tested before it gets released in the Production environment. Ensure all regression tests (RTs) and critical flows work in this environment.
        - **Production Environment** - Once the changes are thoroughly tested, it is promoted into the Production environment.

Create Communication Plan - The Communication Plan ensures that all stakeholders are equally informed of how, when, and why communication will happen. Communication is often a very effective way to solve problems, deal with risks, and ensure that tasks are completed on time. Successful communication plans will identify stakeholders, the information to be communicated, and how this information will be communicated.

- **Identify Technical Requirements for Change Management Solution** - For effective Change Management, define technical requirements to facilitate identification, tracking, deploying and measuring changes. Taking an example, for managing changes in a database environment using Oracle solution, the following are the key requirements:

1. My Oracle Support for Integrated Support
2. Oracle Enterprise Manager Grid (EM12C) provides the management framework and connection to ticketing system

3. OEM 12C Lifecycle Management pack
4. A ticketing tool – for example - Siebel Help Desk for change management which connects into OEM for tracking and managing changes

If this was an E Business application environment, the following are the additional recommendations:

Oracle Application Change Management Pack for Oracle E-Business Suite (Oracle Application Management Suite for Oracle E-Business Suite being a prerequisite).

TA.100 Develop System Management Procedures

Change Management is essentially a workflow where changes go through a lifecycle from logging all the way to the change being released, tested and closed. This section provides best practices to operationalize Change Management procedures using workflows, templates and documentation based on the business requirement:

- **Define Workflows and Templates** - It is important to identify how a change will actually be handled. You need to define the process and procedures for handling change and these should be incorporated into the tools which will handle part of the process. Below are some guidelines specific to IT Change Management:
  - **Define Change Management Workflow** - Define a process flow for your change. Products providing tracking capabilities like Siebel Help Desk will typically have a workflow built in the system and will allow you to modify those to suit your requirements. This process flow will also be integrated with the Approval mechanism to route the change request accordingly. There can be several workflows based on the type of change request (standard, emergency, urgent).
  - **Define Request for Change Templates** - Define a standard template to request for change. These are templates for the requestor to submit when a change needs to be rolled out. The organization can streamline the request by using pre-populated fields and drop-down lists usually provided by the ticketing system (such as Siebel Help Desk) for information such as the type of change, the service affected by the change, and the applicable technology.
  - **Plan Communications (Projected Service Outage)**
    - Communicate changes in advance for the benefit of customers and other stakeholders. One of the key aspects of communicating changes is the Projected Service Outage (PSO) which is used to communicate changes in availability due to scheduled maintenance.
  - **Create Change Models** - Create a repeatable way of dealing with a particular Category of Change which defines specific pre-defined steps that will be followed. Change Models may be very simple, with no requirement for approval (e.g. Password Reset) or may be very complex with many steps that require approval (e.g. major software Release). These models are usually input to the change management support tools in use(such as OEM and Siebel Help Desk) and the tools then automate the handling, management, reporting, and escalation of the process.

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OPERATIONS AND SUPPORT (PS) PROCESS GUIDANCE

PS.010 Audit System

Assets and configurations are required to be evaluated related to business best practices for configuration, security and storage. Compliance standards and audit requirements are defined based on the requirements of the organization. Auditing against compliance standards are operational tasks which should be carried out at a pre-defined intervals to ensure the environment meets the standards.

- **Comply with IT Policies** - Comply with Industry IT Policies by automating real-time IT configuration change management through comprehensive, continuous detection, validation and reporting of authorized and unauthorized configuration change. OEM 12C Lifecycle Management Pack includes a feature set that provides capabilities to comply with Industry IT Policies like SOX,PCI, etc.

  Compliance Management provides the ability to evaluate the compliance of targets and systems as they relate to business best practices for configuration, security, and storage. This will enable organizations to provide the governance to the specific Industry standard they want to comply against.

  Oracle Enterprise Manager 12C (OEM) provides capabilities to create Multiple levels of hierarchies to manage compliance of configuration:
  - **Compliance Framework** - Conceptual “folders” that map configuration standards to real-world structure of compliance frameworks (PCI, COBIT, HIPAA,

CIS, etc)

- **Compliance Standard** - Collection of rules that gets associated to multiple targets
- **Compliance Rule** - Checks/Tests that are performed against specific target types

There is a rich set of out-of-box rules mapped to out-of-box standards and frameworks such as for Exadata configuration, CIS, PCI. To glue all the different compliance standards for particular target types together, we use the compliance framework. This follows the industry standard model to support real-world structures, frameworks like PCI. The frameworks can help the administrators to create rules and standards; the compliance and security officers and auditors can take advantage of the standards and frameworks to manage compliance reports.

For more information on Creating a Compliance Framework, refer to EM Documentation.

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ORACLE SUPPORT SERVICES SUPPLEMENTAL TASK GUIDELINES FOR INCIDENT MANAGEMENT

This component contains OUM supplemental task guidance for managing incidents in an Oracle environment. The guidance complements and extends the baseline OUM guidance and should be used in conjunction with the guidelines in the OUM Task Overviews.

- The first column of the following table provides a complete list of all tasks in this document for which "supplemental" guidelines are provided as well as a quick link to the "supplemental" task guideline in this document.
- The second column provides a link to the OUM Task Overview.

The task guidelines in this document should be used to supplement the guidelines that appear in the OUM Task Overview. The guidance included in this component is focused on on Oracle's recommended approach and best practices for handling Incident Management in an Oracle eco system.

Oracle Support Services - OUM Supplemental Task Guidelines (this document) collapse table | expand table	OUM Task Overview
Technical Architecture	
TA.060 Define System Operations and Management Strategy	TA.060 Define System Operations and Management Strategy
Transition	
TS.040 Design Production Support Infrastructure	TS.040 Design Production Support Infrastructure
Operations and Support	
PS.050 Analyze Problems	PS.050 Analyze Problems
PS.060 Monitor and Respond to System Problems	PS.060 Monitor and Respond to System Problems

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TECHNICAL ARCHITECTURE (TA) PROCESS GUIDANCE

TA.060 Define System Operations and Management Strategy

When preparing the System Operations and Management Strategy for managing your new systems, consider the following when addressing problem identification and resolution, service levels, and metrics and reporting.

- The Service Level details should include the nature of technical support, help desk procedures, service level requirements and reporting.
- The collection of metrics and reporting will be key components for quality improvement.

The primary goal of the Problem Management process is to create a more stable IT service environment by minimizing the adverse impact of errors within the IT infrastructure and to prevent recurring incidents related to these errors. It's important to establish an Incident Management and Problem Management process that eliminates recurring incidents by categorizing the underlying root cause of the problem and then initiating actions to improve or correct the situation. In order to find, analyze and eliminate problems from your environment, you need established processes for detecting problems, determining the root cause of the problem and implementing a solution to the problem. This requires a disciplined set of steps throughout the lifecycle of an incident or problem.

Problem identification and resolution is part of the Incident Management process. The goal of Incident Management is to ensure that normal levels for your IT services are restored as quickly as possible. Then through analysis, establish a quality process that prevents a recurrence of most issues. The strategy needs to be defined in conjunction with service level definitions for your IT service. You will need to establish clear definitions of what constitutes an incident or problem, it's impact to your services, how quickly it must be remedied, how work is assigned, tracked and progressed into your Change Management, Problem Management and Quality Management processes. Your strategy should set acceptable performance metrics that are viewed within both the incident lifecycle and within your process improvement reviews that are part of your Quality Management process.

Your issue management strategy impacts resource planning and your ability to meet services levels. You need to take into account the following components:

- Define roles and responsibilities for Incident and Problem Management.
- Identify key stakeholders and service owners that must be informed and consulted during problem lifecycle.
- Consider two or three categories to group your business services by criticality. Some business services may be "Mission Critical" where an outage may have serious financial impact.
- Establish a definition for an incident. It is suggested that you look at well known IT management practices, such as ITIL.
- Determine how you will describe the severity level for an incident. It is suggested that you adopt well know IT management practices, such as ITIL.
- Decide on response, resolution and escalation times for the combination of service criticality and incident severity.
- Create a definition of priorities to group incidents based upon service criticality and incident severity.
- Target your ratios of incidents that can be self-solved by internal users via automation, solved by first level support (typically the help desk), or solved by a functional or technical expert. Then state how you plan to reach those ratios.

In order to communicate effectively to the management team, to a service owner, or to an end user, you need a means of tracking and reporting issues, their progress and their status. Your management system may be a federation of tools or a central repository that is fed from multiple sources. Consider that end users need to report issues ranging from provisioning problems to functional problems with your services. They may use web forms, email, chat, or a call to a help line. In addition, your technical support team may be using enterprise management tools to monitor critical processes that look for events that meet the criteria of an incident.

A federated system needs a core level of commonality to do reporting against the macro environment, such as a system status web page that people can review prior to opening a service request in one or more of your tracking systems. In addition, you need the ability to extract problem and incident details that can be imported into your Problem Management system for analysis.

A centralized system needs services or methods to receive issues and problems from multiple sources within a defined, acceptable time frame. Enterprise management systems such as Oracle Enterprise Manager often have integration points to create and update incidents within a centralized help desk product such as Siebel Help Desk.

Your issue and Problem Management system must be capable of:

- tracking incidents and problems through the lifecycle – detect, describe, investigate and diagnose, resolve and recover, and close
- capturing enough detail for your Problem Management and quality systems – problem type, severity, impact, duration, root cause, services affected, and more
- managing work assignments and escalations

- clustering or grouping related incidents into a common parent problem
- providing accessible metrics for quality and Problem Management systems

Whether explicitly captured through a web form, or provided by an underlying service, your system should capture the following data types:

- Unique Incident Identifier
- Source of Incident Detection
- Reported By – includes identifying if this was reported by an enterprise management system
- Date Submitted
- Date Last Updated
- Issue Description consisting of short title and long description
- Severity
- Affected Service, Process or System
- Problem Type
- Current Status
- Assignment – current owner of the resolution
- Escalation Status
- Comments (by date/time and author)
- Relationship to Other Incidents or a Parent Problem
- Root Cause Statement
- Solution Link or Description
- Solution Category
- Resolution Date
- Resolution Owner
- Close Date

The strategy for collection and reporting is a the basis for quality control to track adherence to service levels, whether you are meeting your business objectives, what needs to be improved, and how you are progressing with improvements. Incident and Problem Management feeds metrics into the Quality Management process. The metrics look at operational or process efficiency and effectiveness. They should show how well you are meeting your service level response and resolution time for incidents and they should also report on how your processes are maintaining or improving the service levels.

Incident Management goals include:

- Restore service operation as quickly as possible - as measured by time to resolution (TTR).
- Eliminate incidents through problem prevention - as measured by two metrics:
    - Volume of Incidents in a time period
    - Number of Recurring Incidents or known errors in a time period
- Minimize impact of problems - as measured by quality and availability.
    - Mean Time Between Incidents (MTBI) - the rate of incidents measured between the close of the prior incident and the start of the new incident
    - Unplanned down time, or unplanned service interruption should be part of your Service Level Agreements

Incident Management operational efficiency metrics will not only look at your time to resolution, but should also look at key metrics throughout the lifecycle in order to enable your continuous process improvement plans or to be used in formal quality programs.

- Detection
    - Quality of Detected Incidents - number of false alarms vs. impactful incidents should drive changes to your event rules
    - Timeliness of Impactful Incidents - Impactful incidents that are user detected may be candidates for event monitoring
- Record
    - Time to Assignment
    - Time to First Technical Response
    - Number of Escalations

Initial Resolution during Triage - how many user reported incidents, not related to provisioning, are user error
- Investigate and Diagnose
  - Time to Reach Root Cause - Are you capturing the time to diagnose vs. the time to implement the resolution?
  - Time in Wait States - time waiting on vendor, time waiting for other internal resources
- Resolution
  - Actual vs. Estimated Time to Implement
  - Time in Wait State - time from root cause to change implementation
- Close
  - Volume of Self-Solved vs. Vendor Solved
  - Time to Resolution
  - Classification of Closure Decisions (Resolved, Deferred, Closed without Resolution)

Additional reporting should include specific initiatives to improve both the stability of your services and the Incident Management process itself.

- Improvement Description
- Improvement Plan – steps, start and end dates
- Affected Metrics that show success of the improvement process
- Improvement Plan Status
- Improvement Plan Results

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TRANSITION (TS) PROCESS GUIDANCE

TS.040 Design Production Support Infrastructure

The work product for designing the Production Support Infrastructure includes developing the requirements and the processes for your Issue Tracking System and your Configuration Management System. As part of developing the requirements, it's important to understand the relationship between Incident Management, Problem Management and Configuration Management within your support infrastructure. Use the following guidance in conjunction with guidance related to managing production system performance, (that is, PS.050 - Analyze Problems and PS.060 - Monitor and Respond to System Problems).

Incident Management is the process responsible for managing the lifecycle of all incidents. Incident Management's goal is to restore normal service operation as quickly as possible to minimize the impact to the business. The support infrastructure should include systems and processes that can automate the detection, recording, and diagnostic phases of the incident lifecycle. The system also needs to work in conjunction with Problem Management and Change Management systems and processes.

The Product Support process works best when IT Management systems interface along these processes. Consider the following core interaction of Incident Management process with other IT Management systems.

Configuration Management Database (CMDB) stores the configuration items and change history records for your services and systems.

- CMDB needs to receive incident records and problem records linked to configuration items.
- CMDB contains information about configuration items (CI) that may be needed in the Incident Management process. The configuration item information is recorded in the incident record and used during the investigation and diagnosis phase. Comparing the CI record against actual production can yield a potential cause due to unmanaged changes or drift in configuration settings.

Problem Management stores incident record meta-data, analytical results, and actions intended to reduce recurring problems.

- Problem Management Database (PMDB) needs information from the incident containing root cause and solutions in order to identify repeat incidents.
- The help desk needs known problem information and solutions from PMDB that lists work-arounds, and solutions allowing quicker internal resolution of the incident.

Change and Release Management stores records and results of change requests, including risk/benefit analysis.

- Change Management receives change requests as part of the Incident Management flow when a solution is proposed.
- Change Management systems provide a record of recent changes that may be the cause of a new incident. Access to the systems is needed in the diagnose phase.

Enterprise Management (EM) System contains records of monitored events and incidents.

- EM provides detection notification of an event which is classified as an incident.
- EM provides updates on monitored incidents.

Support processes and diagrams show expected flows for incidents, including actions by specific roles and systems used within the processes. Desk manuals for the processes contain definitions required by help desk and support personnel that describe terms such as service criticality, incident, severity, priority, and status. These reference sources should be made available electronically, and should be accessible within the work flow.

The support infrastructure needs interfaces for individuals to report incidents and track the progress of the incident. Your design needs to address how you will receive notification of incidents and updates to the incidents. Will you provide web, mobile, chat, and phone access to create, view and update your incident records?

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OPERATIONS AND SUPPORT (PS) PROCESS GUIDANCE

PS.050 Analyze Problems

Problem management receives input from your Help Desk or Incident Management systems in the form of incident records. The incident records provide details used for problem analysis. Important data used in the analysis include clustering or grouping of incident information along multiple dimensions. Your goal is to improve IT Service Quality through the elimination of problems and by increasing the first time fix rate of incidents that are reported.

Problem records are created if there is a business benefit to find the root cause, which means the resources and costs must support the business impact. Problems are the cause of one or more incidents. Where the Incident Management process is responsible for the quick resolution of the incident, the Problem Management process is responsible for tracking and analyzing the problems that cause the incidents.

Incident data needed for analysis includes:

- Unique Incident Identifier
- Date Submitted
- Issue Description - consisting of short title and long description
- Severity
- Priority
- Affected Service, Process or System
- Problem Type
- Relationship to Other Incidents or a Parent Problem
- Root Cause Statement
- Solution Link or Description
- Solution Category
- Resolution Date
- Resolution Owner

PS.060 Monitor and Respond to System Problems

ITIL describes an incident as an unplanned interruption to an IT service or a reduction in the quality of an IT service. A failure of a configuration item that has not yet affected service is also an incident. For example, if an item within a redundant set of items has failed, such as a mirrored disk, a load balancer, network card, etc., it is still considered an incident due to the potential of reducing service levels.

Incident Management is the process of managing the lifecycle of the incident. The purpose of well executed Incident Management is to restore service operation as quickly as possible with minimum disruption to the business. The goal is to achieve the highest levels of availability and service operation. Incident Management is often paired with Problem Management to prevent recurring incidents as well as manage the volume of incidents.

The lifecycle phases of Incident Management are:

- **Detect** - Looks at conditions and considerations for setting up automated detection based upon events, as well as best practices for handling conditions discovered by people.
- **Record** - Establishes best practices to record, match, categorize and track incidents in order to meet designed service levels as well as contributing to process and quality improvement efforts.
- **Investigate and Diagnose** - Provides guidance on improving your ability to solve incidents related to Oracle products, whether you are self-solving, using social media, or obtaining assistance from Oracle.
- **Resolve** - Describes how you manage implementing solutions within common Change Management practices.
- **Close** - Completes the tracking and Incident Management process with important steps that map into continuous Process Improvement and Problem Management processes.

Detect

Detection is the first step when the incident is identified and notified to the Operations team. Incidents can be detected through automation in the form of event monitoring, or they can come from users or people reporting or logging the incident through your help desk.

Automated detection can be performed by your product's diagnostic framework, by an enterprise management system, or through custom scripts and jobs created by the technical operations team. Most customers will have an enterprise management product, such as Oracle Enterprise Manager, to administer, manage the lifecycle, and monitor the systems in their data center. These same tools have out of the box event monitoring capability as well as providing for user defined event monitoring.

Establishing the rules for converting events into incidents will be specific to your organization. Most organizations will follow a similar approach. The goal is to establish conversion rules that help you focus resources on the most important work in order to maintain a stable, highly available IT Service.

Start with grouping your events into types and severity levels. Example severity levels for a monitored event may be:

- Fatal - Target is down or service is not running.
- Critical - Part of a service is not functional and requires immediate action.
- Warning - Attention is required, but service or area is still functional.

Identify targets that you are monitoring with your IT services. If you are using Oracle Enterprise Manager (OEM), group your targets, not only by type, but by IT service. It is important to know whether a monitored event is affecting a mission critical application rather than an often used, internal application. Adding management packs to OEM will assist in establishing vital relationships between targets.

Establish event types that you will map to targets and severities. A sample set of event types include:

- Availability - service/system down or unreachable
- Performance - service/system not performing to defined level
- Job Status Change - Processes or jobs that may not complete successfully
- Compliance Violation - configuration change no longer matches compliance standard
- Metric alert - monitored metric is outside of defined threshold

Defining Incident Categories helps you map event types, severity, target types and target lifecycle that you use to define whether a detected event is an incident. You also

use the mapping to determine what actions need to be taken and the urgency of the action. Incident categories will reflect some of your event types. For example:

- Availability
- Capacity
- Error
- Job
- Performance
- Security

Record

Recording incidents and the outcome of the resolution process is used to track a wide set of metrics and improvement steps. A primary consumer of the information recorded during the incident resolution process is Problem Management. The goal of Problem Management is to proactively prevent incidents from happening and to minimize the impact of incidents that cannot be prevented. Problem Management relies heavily on the data found in help desks and Incident Management systems. The known errors database is often considered part of Problem Management, thus recording the outcome is key to building a strong knowledge base.

Recording the incident involves gathering data in a form that is useful for future problem prevention. When an incident is logged, it should be initially categorized by service and problem type, and by severity or priority. It needs enough descriptive information to match it against other incidents that may be reported for the same type or matched to known errors or issues. You want your resources focused on the central problem rather than chasing individual incidents. Initial triage is also done at this step for manually detected incidents to determine whether user error is involved or whether the event warrants investigation and resolution.

The main tasks in the Record phase should be:

- Capture incident details automatically, via web form, or through interaction with help desk using phone or chat.
- Confirm severity, problem priority, and problem type.
- Validate issue by capturing symptoms and steps to recreate.
- Establish contact name and method for contact throughout the lifecycle.
- Conduct initial triage:
    - Do skills and resources exist to resolve the problem via the help desk (level 1 support)?
    - Are their sufficient details to diagnose issue and determine root cause?
    - Is this a known problem for which a solution or work-around exists?
- Assign ownership based upon initial triage results>

During the Record phase, self-solve options should be provided to the end user.

Some companies may have more than one system for recording and tracking incidents. For example, Oracle Enterprise Manager provides the ability to manage incidents that would normally be assigned to database administrators and system administrators whose primary functions is in the IT Operations of a data center. Companies may also have rules that determine whether an operational incident needs to be recorded against a central Help Desk system. User detected incidents will likely be logged into a help desk and assigned to a level 1 support person. If a member of IT Operations needs to be involved, the incident may be assigned from 1st level to IT Operations. This would indicate that someone in IT Operations would work out of both a help desk and an Enterprise Management system.

Investigate and Diagnose

Solving incidents should start with a simple plan and some basic understanding of the products in order for you to optimize your time. As an example, since database related error messages can be delivered through any product that uses an Oracle database, it's best to include the product or process information along with the error. Troubleshooting the error is normally best done by filtering the search results based upon the product, version and process (or context) under which that error message was shown. Another example is in the Application space, where Oracle has three product families that provide payroll services and therefore receive tax and regulatory updates on a regular basis. If you are searching on an error or issue related to payroll, you should filter your searches by product, version and process to keep your result list manageable.

The main tasks for the Investigate and Diagnose phase should:

- Verify problem description and steps to recreate.
- Collect facts, including data and files related to the problem.
- Conduct initial investigation:
    - For system detected incidents, did system provide a diagnosis and potential solution?
    - Does the problem and severity require immediate assistance from the vendor or developer?
    - Does the owner have the skills to diagnose and determine root cause?
    - If owner is unable to resolve using internal, community, or vendor support, should owner request vendor or developer assistance?
- Determine the root cause based upon the diagnostic steps.
- Establish proof that issue and root cause are linked.
- Identify one or more solutions to the root cause.

Consider performing the following steps to optimize the path to discovering the root cause for your issue:

- Validate the Incident Description - make sure you have a clear understanding of the incident, especially when reported by end users.
- Restate the incident description in technical terms to assist you in searching for known errors or bugs.
- Incident Cause determination is your initial idea of what the root cause might be. It's your hypothesis to test.
- Incident Cause confirmation is where you validate your hypothesis based upon data and diagnostic tests.
- Solution selection is the step where you are presented with one or more solutions to the root cause.

Investigating incidents for Oracle products have specific, best practices steps depending upon the product, however a common approach and set of tools should be leveraged. Listed below are the core tools you can use.

- Search Product Documentation – useful for error messages related to configuration, setup, and common process mistakes
- Product Diagnostics – Most Oracle products have built in diagnostics for some classes of problems. Product documentation will show you how to enable the diagnostics
- Data Collection – Collecting facts about the incident is critical. Each product has tools to collect data for you. Oracle Database 11g, Fusion Middleware, and Fusion Applications use the Oracle Diagnostic Framework and can write facts into a file system structure known as the Automatic Diagnostic Repository (ADR)
- Oracle Enterprise Manager – The Incident Workbench in OEM provides a guided resolution for many incidents and problem types
- My Oracle Support – The online support portal with access to Oracle's knowledge base, documentation, bug database, and patching database
- Oracle Technology Network and Oracle Forums – publicly accessible tips and suggestions for Oracle products including access to a vibrant community of users

Be prepared to seek vendor or developer help when the severity and priority level of the incident is high and self-solve success is about to cross an acceptable threshold.

Cause determination and justification is a step that ties back into your Problem Management process. During the Investigate and Diagnose phase, you will have formed at least one hypothesis for the root cause of the problem. You would have collected data and found supporting knowledge that helped you prove the hypothesis. Documenting this information allows you to enrich your known errors database that can be used for matching if the issue occurs again or if you choose not to apply a permanent solution to the root cause. You may in fact choose a work around as a solution, which means your incident may reoccur. Having this documented as a known error will save your company time in the future.

To determine the root cause of the problem, you need to establish your hypothesis and test it. This is an iterative step. You may have to reform your hypothesis if the results don't point to a justifiable root cause. Know your environment to help break the problem into probable areas to test first.

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